Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Origin of electrode potential, cell potential

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Origin of electrode potential, cell potential
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Origin of electrode potential, cell potential

1 Origin of electrode potential, cell potential

Study focus: Origin of electrode potential, cell potential. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we will start with a new module on electrochemical equilibrium.

Key details

- In this module we are going to study about electrode potential and cell potential.
- We will derive the Nernst equation.
- Then we discuss the different types of electrodes.
- We will look at reference electrodes and also talk about concentration cells.
- And finally, we will discuss about ion selective electrodes.
- In today's class we will discuss the origin of electrode potential, we will talk about cell potential and we will derive Nernst equation.
- Electrochemical equilibria deals with electrochemistry, which is a branch of chemistry which deals with the interconversion of chemical energy and electrical energy.
- In the galvanic cell, we have chemical energy getting converted into electrical energy.
- It is the energy in a chemical reaction.
- Now the reaction has to be a spontaneous reaction because we are not going to provide any energy from outside.
- The chemical energy has to produce electrical energy.
- And since we are going to produce electrical energy, that is power, there has to be exchange of electrons.
- Therefore, we have to have detox reaction.
- Examples of galvanic cell would be in batteries and fuel cells.
- We convert electrical energy to chemical energy.
- Electrical energy is provided externally and.
- We want to get chemical energy, meaning we want to make a chemical reaction happen.

- We require a non-spontaneous redox reaction in an electrolytic cell.
- We use such cells in electroplating, electro refining, and many other metallurgical processes.
- Whenever we do electrochemical studies, the first thing we require is a redox reaction because we are talking of conversion of chemical energy into electrical energy or electrical energy into chemical energy, and both require an exchange of electrons.
- And one thing we should always remember that oxidation and reduction have to happen simultaneously.
- The second requirement is that of electrodes.
- These oxidation and reduction reactions have to happen at the anode and cathode respectively.
- And the third requirement would be electrolytic conductance.
- The conductance is through the movement of ions.
- We have to remember that electrons cannot move through the electrolyte.
- It's only the ions which are going to cause electrolytic conductance.
- We can also have molten electrolytes.
- Apart from that.
- There is a third category of electrolytes which are called solid electrolytes.
- That is the criteria for an electrolyte.
- What happens is.

Example or application from the lecture

- Suppose we have sodium chloride, we know that when we melt it, then the ions become free to move, and therefore we can use molten electrolyte also in galvanic cells or electrolytic cells.
- For example, we have zirconium oxide when it is doped with Y2O3.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

Oxide ion vacancies are created and this oxide ion can move through the solid electrolyte, therefore it acts as an electrolyte.

Key details

- Since it's allowing for the movement of the oxide ions, we can say that zirconium oxide dope by two or three is also an electrolyte.
- When a metal rod is dipped in a solution of its own ions.
- At the interface there is a potential which is developed and this is known as electrode potential and this happens at the interface of the metal and the solution and it is denoted by capital E.
- The first one is ionization or oxidation the metal rod can dissolve and get changed into Mn plus thereby giving out electrons.
- The ionization reaction is faster than the deposition reaction.

- Just now I mentioned that whenever we have a metal rod dipped in a solution of its own ions, there are two possibilities.
- Either the metal gets ionized or the metal ions take up electrons and yet deposit.
- That is M is becoming Mn plus the N electrons.
- We have M and we have the solution which contains Mn plus.
- When this reaction takes place we know that the metal is getting ionized.
- But what happens to the electrons?
- They are left back on the surface of the electrode, and therefore the electrode acquires a negative charge.
- This at the surface of the electrode we have negative charge, and because there is a negative charge on the electrode it is going to attract the positive ions from the solution.
- We have a positive layer which is getting formed, and because of this we see that at the interface.
- There is an electric double layer which is getting formed.
- This electric double layer is responsible for the electrode potential and it is known as the hellholes electric double layer.
- metal rod and it is dipped in a solution of Mn plus.
- But this is case two where the deposition reaction is faster, meaning the reduction reaction is happening faster.
- And get deposited.
- But where are the electrons coming from?
- They are coming from the metal rod.
- Again this electrical double layer is responsible for the electrode potential at the interface and this electrical double layer is known as the.
- And molds electrical double layer.

Example or application from the lecture

- Let us now discuss the origin of electrode potential.
- Let us take up a case where.
- Now suppose the first reaction is faster where wherein the ionization reaction is faster.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

or the reduction reaction that is deposition happens to be the faster reaction.

Key details

- In both cases we understand that there's going to be an electrical double layer which is formed at the interface between the metal rod and the solution and this is known as the Helmor's electrical double layer and responsible for the electrode potential which we see in this system.
- Whether the oxidation reaction will happen faster or the reduction reaction will happen faster is going to depend on the nature of the metal.

- If the metal has greater tendency to undergo oxidation, we find that ionization will be the faster reaction and if.
- The ions have a greater tendency to undergo reduction, then we find that the deposition reaction will be happening at a faster rate.
- So that means it clearly depends on the nature of the metal, and it also depends on the concentration of the ions which are being used, it depends on the temperature.
- And if gases are involved on the pressure.
- If we want to compare electrodes, then we need to define a new quantity which is known as the standard electrode potential.
- It is defined as the potential developed at the interface of the metal in the solution.
- When the metal is in contact with the solution of its own ions, but now we have placed a restriction that the concentration should be unity and the temperature is now fixed at two ninety eight Kelvin and in the.
- In case there are gases involved, then the pressure of the gas also has to be maintained at one atmosphere pressure.
- But single electrode potentials cannot be measured.
- What we can measure is the potential difference.
- Therefore, we need to form a cell where two electrodes are coupled together and such a cell would be called as an electrochemical cell.
- And therefore we see.
- We know that in a Daniel cell what is it?
- Zinc is dipped in a solution of zinc sulphate, copper is dipped in a solution of copper sulphate.
- Zinc rod is dipped in a solution of zinc sulfate.
- We always write the anode first.
- Where oxidation happens, the anode is written first.
- Between zinc and copper we find that the zinc electrode will undergo oxidation.
- The value in this parenthesis is the concentration of the zinc ions.
- So anode is written first and this single line over here represents the phase separation.
- So zinc is solid and Zn two plus is in the liquid form, therefore there is a phase separation shown by the single line.
- Again is separated by a single line, but and this is where this is the cathode.
- So we have the anode first and then we write the cathode, and in between we can see these double lines.
- This represents the salt bridge.
- What we have to remember is we cannot write.
- Zinc Zn2 plus and salt bridge copper and then CU2 plus.
- Because the salt bridge is connecting the solutions.
- Therefore, this is the representation and this is known as cell notation.
- The difference in electrode potentials of the electrode is something which we can measure and that is known as the cell potential.
- Potential and it is denoted by e cell.
- Again, cell potential is going to depend on the concentrations of the solutions used and also on the temperature and also on the pressure as well as the electrodes which have been used.

- So therefore, in order to.

Example or application from the lecture

- Suppose we take the example of Daniel Cell.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

compare we need to have a standard value which is known as the standard electrode potential and e cell depends on the concentration temperature and partial pressure of the gases involved.

Key details

- Cell potential is.
- Because we know the cell is made of two electrodes, one electrode is the anode electrode, the other electrode is the cathode electrode.
- And also I mentioned that anode is written first and then that is on the left hand side and cathode is written on the right hand side of the salt bridge.
- Because it represents the driving force for the cell reaction to take place.
- What does it tell us that for any spontaneous reaction we know that delta G has to be negative and therefore E cell should be positive.
- And if the reaction is non-spontaneous, then delta G will be positive and therefore the E cell has to be negative.
- The standard reduction potentials have been listed in a table in ascending order.
- This is to help us predict the electrochemical behavior of an electrode electrolyte system.
- This series is known as the electrochemical series.
- Most often we find that it is the standard reduction potentials which have been used and such an arrangement is referred to as the electrochemical series.
- This series is going to be very informative.
- A negative value of standard reduction potential indicates oxidation tendency while a positive value is going to indicate a reduction tendency.
- Whereas copper we find that.
- CU2 plus taking up electrons giving you copper that is the reduction potential is plus 0.34.
- Hence the value is positive.
- Electrochemical series lists the standard electrode potentials which are measured under standard conditions of unit concentration two ninety eight Kelvin temperature and one atmosphere pressure.
- But most of the times we do not deal with standard conditions.
- Now the reaction at the electrode is the reduction reaction, so we write it as Mn plus N electron is equal to M because we are talking of reduction potentials.
- Written as minus delta G is equal to W max.
- The system we are talking about here is a metal rod dipped in a solution of its own ions.

Example or application from the lecture

- E cell is given by electrode potential of the right hand side electrode minus the electrode potential of the left hand side electrode or in other words we can say e cathode minus the e anode.
- because the change in free energy for any reaction any system tries to go to minimum free energy so the change in free energy is given by minus nf ec.
- The of the two electrodes, and that will be the driving force for the reaction to take place, the difference in the free the electrode potentials will decide the reaction the they decide the reaction and in other words thermodynamically that is expressed as delta g is equal to minus nfe c let us talk about the electrochemical series.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

For any electrical chemical system, the maximum work done is given by the total charge available multiplied by the energy available per unit charge.

Key details

- The total charge available here is equal to the number of moles of electrons exchanged in the redox reaction, that is n we can see here, multiplied by the charge carried per mole of electrons, and that is f.
- Therefore, now the W max becomes equal to N F.
- multiplied by E.
- That is the total charge available is Nf and the energy available per unit charge is E, therefore W max is equal to N F E and therefore delta G becomes equal to minus N FE by substituting in this equation.
- Under standard conditions, delta G is delta G naught that is equal to N minus of Nfe.
- The reaction we are talking about is Mn plus N electron is giving you M.
- What is q?
- It is the reaction quotient.
- We know what is delta G naught and we also know Q.
- In this equation, we are going to substitute, we get minus of Nfe is equal to minus Nfe naught plus R T log M over Mn plus.
- that is 96500 coulombs per mole.
- For pure substances we take M concentration equal to one.
- Substituting for R 8.314 joules per Kelvin per mole, temperature as two ninety-eight Kelvin, F is nine six five zero coulombs per mole.
- multiplied by 2.303 then we get the value 0.0591.
- This is for a single electrode.
- The only thing which is going to change is the reaction taking place.
- Q value is concentration of the.
- Products raised to the respective stechometric coefficients divided by concentration of the reactants raised to the respective stepometric coefficients.
- Delta G remains the same.

- It is minus N F E and Delta G naught is going to be minus N F E naught.

Example or application from the lecture

- Total charge is N into F and the energy available per unit charge is will be electrode potential because electrode potential is given by energy per unit charge.
- That is if you if the unit is volts then that is given by joules per coulomb and that is the electrode potential which is given by E.
- A thermodynamic equation which is going to react which relates the reaction quotient and the decrease in free energy is given by delta g is equal to delta g0 plus r t log q.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 25:00–29:59

cell emf and the standard cell emf values so dividing by minus nf and of course we have to substitute for q the nonst equation at 298 Kelvin now is equal to e naught minus 0.0591 by n log of.

Key details

- C raised to the power of small c d raised to the power of small b divided by a raised to the power of small a and b raised to the power of small b.
- and n is the number of electrons transferred and e0 is the standard EMF of the cell.
- Now in the next class, we are going to discuss about the different types of electrodes.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Origin of electrode potential, cell potential transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Origin of electrode potential, cell potential transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Origin of electrode potential, cell potential transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Origin of electrode potential, cell potential transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00002	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00003	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00005	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

C Revision and retrieval practice